

William Jackson

Senior Software Engineer – Vue Node (Express/NestJS) AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience delivering **Node.js 8–20**, **Vue.js 2–3**, **TypeScript 3.x–5.x**, and cloud-native web platforms across the **SaaS, enterprise software, and data-driven product industry**, with a strong record of building secure APIs, responsive frontends, and scalable backend services. Strong background in **microservices, REST APIs, GraphQL, PostgreSQL, Redis, AWS, Docker**, and **CI/CD**, including modernization of legacy monoliths into maintainable service-based architectures. Known for resolving production bottlenecks, stabilizing unreliable releases, improving deployment speed, and shipping new platform features that increased performance, reliability, and developer productivity.

SKILLS

- **Frontend: Vue.js 2–3, Nuxt.js 2–3, JavaScript ES6+, TypeScript 3.x–5.x**, HTML5, CSS3, Sass/SCSS, Tailwind CSS, Bootstrap, Vuex, Pinia, Axios, Webpack, Vite
- **Backend & APIs: Node.js 8–20, Express.js 4–5, NestJS 8–10**, REST API, GraphQL, JWT, OAuth2, OpenID Connect, RBAC, WebSocket, Microservices, Server-Side Rendering, API Gateway Patterns
- **Data, Messaging & Search: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, DynamoDB, Kafka, RabbitMQ, AWS SQS/SNS, Elasticsearch, OpenSearch, Redshift, BigQuery**
- **Cloud & DevOps: AWS (EC2, ECS, EKS, Lambda, API Gateway, S3, CloudFront, RDS, DynamoDB, ElastiCache, EventBridge, Step Functions, IAM), Azure, GCP, Docker, Kubernetes, Helm, Terraform, CloudFormation, GitHub Actions, Jenkins, Argo CD, GitOps**
- **Observability, Quality & Security: Datadog, Prometheus, Grafana, CloudWatch, ELK/EFK, OpenTelemetry, Jest, Mocha, Chai, Vue Test Utils, Cypress, Playwright, Pact-style contract testing, Rate Limiting, Audit Logging, Auth0, Okta, AWS Cognito**

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Led architecture and delivery of a multi-tenant **Node.js 16–20** and **Vue.js 3** platform that unified customer onboarding, billing workflows, and operational analytics across several enterprise product lines.
- Designed and implemented modular backend services with **NestJS, Express**, and **PostgreSQL**, replacing tightly coupled legacy modules and reducing average API response time by **38%** under peak business traffic.
- Built high-visibility frontend experiences in **Vue.js 3, TypeScript**, and **Pinia**, improving dashboard rendering speed by **42%** and significantly reducing state synchronization defects across complex user workflows.
- Migrated older frontend modules from **Vue.js 2** with **Vuex** and legacy bundling into **Vue.js 3** with **Vite**, eliminating inconsistent build behavior and cutting local startup time by roughly **55%**.
- Introduced **GraphQL** for selected reporting and account management workflows where REST payloads had become too chatty, reducing over-fetching issues and improving page interaction latency for power users.
- Reworked authentication and authorization flows using **OAuth2, OpenID Connect, JWT**, and **RBAC**, closing several permission edge cases that had caused support escalations in shared-account enterprise environments.
- Implemented event-driven processing with **Kafka, SQS/SNS**, and **Redis** to decouple notifications, billing recalculations, and audit activities, which improved resilience during traffic bursts and release windows.
- Solved a recurring memory leak in long-lived **Node.js** worker processes by tracing unbounded object retention in background job handlers, then refactoring retry logic and queue acknowledgement flow to stabilize runtime behavior.
- Built reusable CI/CD pipelines with **GitHub Actions, Docker**, and **Terraform**, enabling safer deployment promotion, better environment parity, and a measurable **60%** reduction in manual release coordination effort.
- Containerized core services and deployed them through **AWS ECS** and later **EKS**, which improved rollout control, health monitoring, and rollback safety for business-critical services with strict uptime expectations.
- Added observability with **Datadog, Prometheus, Grafana**, and **OpenTelemetry**, allowing the team to identify slow queries, failed queue consumers, and noisy endpoints before they escalated into customer-facing incidents.
- Partnered with product and support teams to deliver new subscription, reporting, and tenant-management features while maintaining strong release quality, helping improve successful first-release delivery by **30%**.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Built customer-facing web applications using **Node.js 12–16**, **Vue.js 2–3**, and **TypeScript**, supporting sales operations, internal workflow automation, and data-heavy reporting across a growing SaaS product suite.
- Developed secure REST APIs with **Express.js**, **PostgreSQL**, and **Redis**, handling account lifecycle operations, permission-sensitive resources, and asynchronous processing for back-office and client-facing modules.
- Delivered reusable Vue component libraries for forms, tables, and role-aware navigation, which improved development consistency and cut duplicate UI effort across multiple product teams.
- Migrated selected backend modules from a monolithic service into smaller API domains, reducing deployment risk and allowing independent scaling for reporting, user management, and notification workloads.
- Resolved a persistent race condition in asynchronous invoice generation by isolating shared update paths and introducing idempotent processing with queue-backed retry controls, which significantly reduced duplicate records.
- Implemented background processing with **RabbitMQ** and **Redis** for export generation, email delivery, and scheduled sync tasks, improving responsiveness of the main application during heavy user activity.
- Improved search and filtering capabilities with **Elasticsearch**, allowing users to query larger operational datasets faster and reducing time spent on manual report preparation for account teams.
- Strengthened application security by introducing **JWT** token validation hardening, route-level authorization controls, audit logging, and API rate limiting for sensitive administrative functionality.
- Helped modernize frontend delivery by moving legacy webpack-heavy views toward cleaner **Vue.js** module boundaries and more maintainable state management patterns ahead of broader platform upgrades.
- Created automated quality gates with **Jest**, **Mocha**, **Cypress**, and API contract tests, which improved confidence in releases and reduced regression issues reported after deployments by **28%**.
- Collaborated closely with QA and business stakeholders to ship new workflow features, improve stability, and raise overall release predictability while supporting an expanding enterprise customer base.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed internal and customer-facing applications using **Node.js 8–12**, **Vue.js 2**, **JavaScript**, and **MySQL**, supporting order management, reporting workflows, and business process automation.
- Built and maintained backend APIs with **Express.js** and SQL-based data access layers, focusing on reliable CRUD behavior, transactional integrity, and practical performance tuning for high-usage modules.
- Created dynamic interfaces with **Vue.js 2**, **Vuex**, and reusable UI patterns that made legacy browser-driven workflows easier to maintain and more intuitive for operations users.
- Solved slow report generation issues by rewriting inefficient SQL joins, adding indexes, and caching repeated lookups with **Redis**, which improved report execution time by nearly **45%** for large datasets.
- Introduced structured error handling and centralized logging for API services, making production issues easier to diagnose and reducing repeated support effort on intermittent failures.
- Implemented file import and export pipelines for CSV-heavy operational workflows, adding validation, partial failure reporting, and retry-safe processing for business-critical daily tasks.
- Supported gradual migration of older server-rendered pages into component-based **Vue.js** screens, improving maintainability while allowing the business to continue shipping without disruptive full rewrites.
- Worked on deployment automation with **Jenkins**, **Docker**, and scripted environment configuration, reducing manual release steps and improving consistency between staging and production environments.
- Addressed cross-browser and state synchronization bugs in complex admin screens by simplifying component communication patterns and tightening client-side validation logic before submission.
- Partnered with senior engineers and analysts to implement new features quickly, fix production issues responsibly, and improve system reliability in an environment with frequent business change requests.

- Contributed to web application development using **Node.js 6–8**, **JavaScript**, **Vue.js 2**, **HTML5**, **CSS3**, and **SQL Server**, supporting line-of-business systems and internal administration tools.
- Built foundational API endpoints, form-driven interfaces, and reporting screens while learning practical software design, debugging discipline, and maintainable implementation patterns in a production team.
- Assisted with enhancement of legacy modules by introducing cleaner component structures and reusable client-side behaviors, reducing duplicate code and making small features faster to implement.
- Resolved recurring validation and data formatting issues between frontend forms and backend APIs by standardizing payload handling rules and improving defensive checks on both layers.
- Supported migration of selected jQuery-heavy pages into early **Vue.js 2** components, giving the team a more maintainable front-end path without requiring an immediate full platform rewrite.
- Wrote SQL queries and stored procedure updates for operational reporting, fixing inaccurate aggregations and improving data consistency for business users who relied on daily summary dashboards.
- Helped troubleshoot release issues by reviewing logs, reproducing bugs in test environments, and documenting root causes so senior engineers could apply permanent fixes more efficiently.
- Improved application stability by adding safer error messaging, basic unit coverage, and deployment checklists that reduced avoidable rollout mistakes during routine changes.
- Gained hands-on experience with **Git**, release coordination, and collaborative delivery practices while contributing reliably to feature work, bug fixes, and support-driven engineering priorities.

EDUCATION

Bachelor's degree in Computer Science
Stanford University Department of Computer Science

2010 to 2014

- Built strong foundations in **algorithms**, **databases**, and **software engineering**, applying coursework through practical team projects and production-style system design.

CERTIFICATIONS

- **AWS Certified Developer – Associate — 2024**
Validated hands-on capability in building, deploying, and troubleshooting secure application workloads on **AWS** using modern development and integration practices.
- **OpenJS Node.js Services Developer (JSNSD) — 2024**
Demonstrated proficiency in designing and developing **Node.js** services and APIs, including service architecture, routing, security practices, and integration patterns for scalable backend systems.
- **Vue.js Certification — 2024**
Validated practical frontend development skills using **Vue.js**, including component-based architecture, state-driven UI design, reactivity, routing, and modern single-page application development practices.